Route-Aware Computational Selection of Redox Motifs for Preformed Polymers

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Internal working draft. Pending calculations and validation are identified explicitly.

Abstract

Selecting a redox motif for a preformed polymer requires useful electrochemistry, feasible attachment and acceptable behavior among neighboring grafts. We link these assessments through exact precursor and product identities, requiring every selection criterion to pass before ranking candidates. An unfitted retrospective benchmark of 39 potentials in acetonitrile gives a mean absolute error of 0.247 V and shows systematic and redox-step-dependent residuals. Defined base-assisted reactions show that a specified base adds a common grafting free-energy term without changing within-base rankings; the computed DBU proton-capture term differs from a MeCN pKa estimate by 22.8 kJ mol^{-1}. Matched five-repeat phthalimide controls give adjacent and one-spacer free-energy nonadditivities of -5.17 and -11.39 kJ mol^{-1} within finite two-conformer sets. Both spacings also permit close-core local minima, so an intervening repeat does not enforce spatial isolation. An exact enumerated PMDI design connects molecular potentials and precursor evidence to composition-dependent capacity ceilings of 61–125 mAh g^{-1}, while selectivity, accessible utilization and polymer behavior remain unresolved. Together, these results establish separate, testable roles for molecular screening, reaction conditions and polymer occupancy in a route-aware assessment.

Keywords: redox-active polymers; molecular screening; grafting thermodynamics; molecular identity; conformer ensembles; acetonitrile; density functional theory; chemical provenance; multiobjective selection.

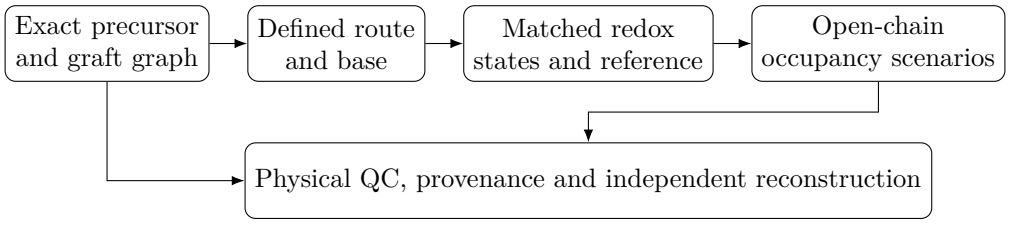


Figure 1: Workflow overview. Molecular redox properties, route-dependent grafting thermodynamics and neighboring-site effects are separate outputs linked by exact chemical identity. Execution correctness is evaluated independently of experimental accuracy. Every selection criterion is assessed separately for the same exact graft identity and declared conditions.

1 Introduction

Redox-active polymers and colloids provide charge-storage architectures that can reduce transport of active groups through separators. Aqueous polymer flow batteries and nonaqueous redox-active

colloids establish distinct realizations of that approach.[1, 2] Kim and co-workers further demonstrated redox-matched insoluble polymer beads coupled to soluble mediators.[3] For an already formed support, selecting a redox group requires both a useful electrochemical response and a feasible reaction at an accessible polymer site.

High-throughput calculations have explored quinone redox couples and databases have organized redox-active chemical space.[4, 5] Such efforts motivate systematic design, while computational electrochemistry identifies reference, solvation and charge-state errors that complicate transferring potentials between chemical environments.[6–8] A potential of a free parent molecule cannot alone establish the behavior of an exact graft. Attachment introduces precursor access, selectivity and reaction conditions; neighboring grafts introduce additional conformational and electrostatic response. Reactive-handle density, conversion, sequence spacing and spatial contact must therefore remain distinct.

Self-association is already a demonstrated design concern for diimide electrolytes. Wiberg and co-workers combined aqueous NDI flow-battery experiments with charge-state-dependent dimer calculations, identifying electrolyte composition and steric substitution as relevant design variables.[9] These findings motivate evaluating neighboring-site effects within a declared molecular and electrolyte context.

General workflow systems already support provenance, failure handling and large quantum datasets.[10, 11] The chemical question addressed here is how attachment conditions and neighboring grafts change the assessment obtained from molecular redox screening. We examine this question through exact-identity links among molecular redox response, base-defined attachment and conditional neighboring-site effects (Figure 1). A retrospective benchmark characterizes molecular prediction errors, balanced reaction cycles isolate the role of the base, and matched occupancy controls distinguish conditional grafting from isolated pair association. An enumerated PMDI example then tests which of these observations supports a material decision. The declared molecular families and sampled conformer sets define the scope of this focused study.

2 Methods

2.1 Molecular identity, candidate generation and attachment routes

We model post-functionalization of a preformed polymer bearing reactive benzyl chloride sites. A 4-methylbenzyl chloride fragment provides a local capped-site reference. Its grafted products represent a polymer attachment site; the full support, crosslinks and graft distribution require separate polymer models.

Systematic designs vary declared viologen, pyromellitic-diimide and naphthalene-diimide cores, attachment linkers, terminal substituents and individual ring substitutions. Exact stereochemical graphs and precursor/product pairs are deduplicated, while every generating recipe is retained. Charge states and multiplicity hints are checked for electron-count consistency. A declared redox ladder remains a computational hypothesis until its states pass the appropriate property-specific checks. Molecular weight and nominal gravimetric capacity, $Q = nF/(3.6M)$, are bookkeeping descriptors; they do not establish accessible electron count, polymer capacity or synthetic feasibility.

The attachment model specifies a charge- and atom-balanced precursor/product pair and a uniquely mapped new heavy-atom bond from each verified upstream design. Evidence is bound to that exact identity, so a precedent for a different product cannot authorize the design. A prospective route gate requires documented precursor access, attachment selectivity, defined reaction conditions, a balanced route, redox-core retention and source evidence. It distinguishes proceed, hold and exclusion decisions. Family-level precedents are retained with their scope and

are not automatically promoted to exact-candidate synthesis evidence. Historical calculated properties and explicitly scoped method controls remain available even when a prospective synthesis decision is unresolved.

2.2 Admission across all required dimensions

A candidate is eligible for ranking only after every required criterion passes in its declared context. The criteria cover operating redox window, material capacity, state stability, intended charge transfer, practical attachment, defined-route thermodynamics, unwanted motif association, unwanted intersite electronic coupling, occupancy sensitivity and polymer accessibility. A failed criterion excludes the design within that route or operating context. Missing evidence or an undeclared acceptance threshold leaves it pending. Route calculations use a reaction context; operating properties share a separately declared material and electrolyte context. Exact product identity is required across the joined records.

For a numerical criterion, the entire qualified uncertainty or scenario interval must lie within the accepted bounds. An interval crossing a boundary remains unresolved. Parent-benchmark residuals are not assigned as candidate-specific prediction intervals. The current material-capacity, charge-transfer, motif-association and intersite-coupling cutoffs remain undeclared; the model operating window and nonpositive defined-route free-energy screen are explicit conventions rather than experimentally validated universal thresholds. Pareto comparison is restricted to candidates that pass every criterion, with the two electrode pools treated separately. No weighted score supplies a missing pass.

2.3 Electronic structure and solution free energies

Geometry-level and native-Hessian calculations use r^{2}SCAN with D4 dispersion and SMD acetonitrile [12–14]. The geometry-level basis is def2-SVP for neutral and cationic states and def2-SVPD for anions. Active electronic energies use ωB97M-V/def2-TZVPD on the selected geometry, with the same solvent model.[15–17] Density fitting uses the implementation-default auxiliary bases with `only_dfj=False`; the hybrid calculations therefore fit both Coulomb and exact-exchange contributions. PySCF 2.14.0 and GPU4PySCF 1.8.1 are the validated implementations for the present native-thermal protocol.[18, 19] Supporting electrolyte ions are not implicitly introduced into these bare-solute calculations. Explicit ion-pair models are identified separately.

The reported molecular free-energy model is

$$G = E_{\mathrm{SMD}}^{\omega\mathrm{B97M-V}} + G_{\mathrm{nuclear}}^{\mathrm{qRRHO}} - RT\ln(2S+1). \tag{1}$$

The nuclear partition uses the solution geometry-level DFT Hessian. Rigid translations and rotations are excluded by a mass-weighted geometric projection, rather than by deleting all frequencies below a numerical threshold. The xTB 6.7.1 thermodynamic kernel is used only as a qRRHO integrator for the explicit DFT internal-mode list.[20, 21] Frequencies are unscaled. Modes between -5 and 0 cm^{-1} are accepted only within the stated minimum tolerance and enter the partition as $|\nu|$; more negative modes fail. The rotational symmetry number is determined by xTB from the exact geometry. The primary rotor cutoff is 50 cm^{-1}. Cutoffs of 25 and 100 cm^{-1} provide sensitivity estimates. This use of the thermodynamic kernel does not substitute a semiempirical Hessian for a DFT Hessian. The spin factor assumes an unsplit ground multiplet; unresolved state ordering remains a separate admission issue.

Thermodynamic acceptance requires the largest absolute Cartesian gradient component no larger than 4.5×10^{-4} hartree bohr^{-1}, Hessian asymmetry no larger than 10^{-6} in the stored

atomic-unit matrix, and a lowest internal frequency no smaller than -5 cm^{-1}. The complete Hessian columns for atom 0/y and the first oxygen/z coordinate are tested at 0.0005 and 0.001 bohr; iron or the final atom is used when oxygen is absent. Each sampled column must meet a 1% relative-norm error threshold. This tests two columns, not the full Hessian. The absolute spin-contamination threshold is $|\langle S^2\rangle - S(S+1)| \leq 0.1$. Full integration-grid response, finite numerical values and spin diagnostics are checked separately. SCF convergence alone is insufficient for thermodynamic admission. Missing thermal or electronic quantities remain missing. Geometry optimization uses geomeTRIC.[22] The charge-dependent geometry/Hessian basis is a protocol imbalance. Five benchmark neutral partners were separately reoptimized with def2-SVPD and evaluated with a derivative-checked Hessian at that level. Their actual alternative protocol identities remain separate from the primary certificate. Existing anion and Fc records are held fixed to separate nuclear and geometry-induced active-energy effects; this is a sensitivity test rather than complete basis convergence.

2.4 Redox reference and retrospective accuracy assessment

For a one-electron oxidized/reduced pair, potentials are calculated from $G_{\mathrm{ox}} - G_{\mathrm{red}}$ and referenced to a matched ferrocene/ferrocenium calculation. The experimental comparison uses exact free-parent identities and documented acetonitrile measurements with direct or source-defined reference conversion. Lower-tier measurements and excluded events are retained in the source catalog.

The frozen retrospective panel contains 39 potential endpoints from 30 molecules. It includes all currently sourced strict endpoints and is not a held-out test set. Independent reviewers reconstruct state free energies, the reference and potential differences from backing records. Statistics use unrounded calculated potentials, retaining family, redox-step and original-study labels. Molecule- and study-cluster bootstrap summaries use 2,000 replicates and seed 20261021; intervals require at least five clusters. These summaries describe this panel and are not calibrated prediction intervals for new structures. The primary statistics use no fitted offset or outlier removal. Post hoc endpoint-, molecule- and study-held-out offsets are reported separately, using either a single training-set mean residual or separate means for the two redox steps. All endpoints of a held-out molecule or study are excluded together. Because the residual structure motivated these analyses, they are internal diagnostics rather than prospective validation.

2.5 Defined grafting reactions and standard states

For an N–H or O–H precursor, the local reference reaction is

$$\mathrm{RCl} + \mathrm{HNu} \rightarrow \mathrm{RNu} + \mathrm{HCl}. \tag{2}$$

Pyridyl quaternization is treated as a distinct reaction class with the appropriate precursor and chloride charge. The neutral-substitution reference is not interpreted as the actual free energy of a synthesis that includes a base. For a defined base B, we additionally evaluate

$$\mathrm{RCl} + \mathrm{HNu} + \mathrm{B} \rightarrow \mathrm{RNu} + \mathrm{BH}^+ + \mathrm{Cl}^-, \tag{3}$$

$$\mathrm{RCl} + \mathrm{HNu} + \mathrm{B} \rightarrow \mathrm{RNu} + [\mathrm{BH}^+ \cdots \mathrm{Cl}^-]. \tag{4}$$

The separated-ion and contact-pair representations are alternative thermodynamic models, rather than an assumed equilibrium population or a precipitation model. Each reaction is checked for atom and charge balance.

The native thermal translation convention is ideal-gas 1 atm. Reaction sums are converted to a 1 M solute standard state by adding $\Delta n RT \ln(RTc^\circ/p^\circ)$. The correction cancels for the

neutral reference and separated-ion routes and is approximately $-7.93\ \mathrm{kJ\,mol^{-1}}$ for the contact-pair route at 298.15 K. The DBU/DIPEA pilot computes each neutral base, protonated base and protonated-base/chloride contact pair. Three geometry seeds are evaluated per new species, and thermochemistry is applied to the selected electronic-energy minimum. Reused component values retain their original source hashes. The difference between paired and separated representations is a model sensitivity, not an uncertainty interval.

A separate external term check compares $\mathrm{B} + \mathrm{HCl} \rightarrow \mathrm{BH^+} + \mathrm{Cl^-}$ with the author-maintained MeCN basicity collection and recommended acid table.[23–25] The conversion is $\Delta G^\circ = -RT\ln(10)[\mathrm{p}K_a(\mathrm{BH^+}) - \mathrm{p}K_a(\mathrm{HCl})]$. Base identity is matched by the exact neutral molecular graph; analogues are not substituted. This comparison uses 298.15 K, while exact source temperature and activity conventions remain under review. HCl cancels from the full all-DFT grafting cycle. Replacing only its capture term therefore tests sensitivity to a term and shared-error cancellation; it does not calibrate the complete reaction.

2.6 Finite-polymer construction and conditional occupancy cycles

Oligomer construction preserves an open carbon backbone, the intended graft bond and declared stereochemistry. Monomers, dimers and trimers are included in the original interaction panel; no tetramers are included. GFN2-xTB/ALPB(acetonitrile) warm starts provide conformational seeds for subsequent DFT evaluation.[26, 27] Starting pose labels do not determine the final contact class. XYZ audits independently inspect composition, bonding, stereochemistry and severe clashes.

The matched spacing control uses a five-repeat backbone with reactive sites 2, 3 and 4 and plain styrene end-padding units at positions 1 and 5. The six occupancy graphs are empty, singly grafted at sites 2, 3 or 4, and doubly grafted at 2/3 or 2/4. Unreacted benzyl chloride and unfunctionalized styrene units remain chemically distinct. The graft is a neutral phthalimide method control. The initial search uses seed 20261020, four embedded structures and two xTB seeds per starting mode, with two selected DFT poses per graph giving twelve electronic calculations. Unbiased starts are supplemented by open and slipped starts for double grafts. One declared backbone configuration is used; it is not a model of an atactic sequence ensemble. Fully occupied short oligomers represent dense local scenarios; these calculations assign no experimental loading distribution or occupancy probability.

The neighboring-site free-energy nonadditivity is

$$\Delta\Delta G_{ij} = G_{ij} + G_{00} - G_i - G_j. \tag{5}$$

This quantity is also the difference between grafting the second site in the presence and absence of the first graft when the same reagent and byproduct terms are used. It includes changes in the backbone, conformation and solvent response and is not a pure pendant-core pair interaction. Electronic $\Delta\Delta E$ and thermodynamic $\Delta\Delta G$ are kept separate. An absolute conditional grafting free energy additionally includes the defined base and byproduct terms.

For separate, intact molecular segments, frozen-fragment energies distinguish interaction from deformation:

$$E_{\mathrm{int}} = E_{AB} - E_A^{AB} - E_B^{AB}, \tag{6}$$

$$E_{\mathrm{def}} = (E_A^{AB} - E_A^0) + (E_B^{AB} - E_B^0). \tag{7}$$

Their sum equals the segment association energy referenced to relaxed isolated segments. This partition avoids cutting covalent bonds or adding new caps. It nevertheless contains the full segments, is not counterpoise corrected, and does not establish an association free energy or electronic coupling.

2.7 Evidence and execution controls

Every reported field points to exact inputs, numerical settings and immutable backing records. Physical checks govern acceptance separately from scheduling success. Durable geometry checkpoints remain unqualified until normal convergence, topology, spin and thermodynamic checks pass. Independent GPU reservations and bounded CPU thread pools govern parallel calculation. Numerical replay, synthetic throughput, CPU concurrency and actual quantum restart controls are reported in the Supporting Information; these test execution or numerical equivalence rather than experimental accuracy.

3 Results

3.1 Systematic designs and benchmark coverage

The declared enumeration generates 1,593 recipes and 837 distinct designs: 459 viologens, 243 NDI derivatives and 135 PMDI derivatives. This pool concentrates on reduction-active motifs and includes two related diimide cores. BQ/AQ and oxidation-active chemistry have historical examples but lack equivalent systematic coverage; shielding and linker rigidity also need deliberate matched comparisons. These boundaries define a focused study rather than a comprehensive chemical-space search.

The historical evaluation comprises 75 candidates and the same 75 reaction identities, with 72 unique precursors. It is distinct from the enumeration, which itself submitted no new full-redox jobs. Exact graph matches can reuse historical calculations with matching state and protocol provenance. Their union with one method control contains 879 distinct graphs. All 878 candidate designs remain pending under the all-axis profile; no complete material shortlist is inferred from the molecular descriptor front. This count primarily reflects incomplete criteria and evidence, rather than demonstrated failure of 878 chemistries. The exact PMDI-ethyl assessment below illustrates how the route and material requirements constrain a candidate with existing molecular results.

The retrospective free-parent benchmark includes 39/39 sourced potentials from 30 molecules and seven studies, with MAE 0.247 V, RMSE 0.279 V and bias +0.190 V (Figure 2). Molecule-cluster 95% bootstrap intervals are [0.213,0.286], [0.245,0.313] and [0.128,0.257] V, respectively. Seven unequal study clusters provide weakly calibrated intervals, reported separately in the SI. An independent reconstruction contains 111 checks with no discrepancies. The final PMDI second-reduction endpoint is included; its alternative triplet lies 0.707 eV above the selected singlet electronically, while the alternative thermal spin gap remains unresolved.

Coverage differs materially from the design space. Twenty-two endpoints are quinones, which are absent from the systematic enumeration. The imide subset contains nine endpoints from six molecules in one study, including three PMDI derivatives and three phthalimides; viologen contributes one molecule, and there is no NDI endpoint. These results define accuracy for the measured free parents and do not validate 837 grafted designs or polymer environments.

3.2 Reference bias and charge-state errors are distinguishable

All 30 first-step errors are positive, with mean +0.266 V. The nine second-step errors have mean −0.064 V and MAE 0.185 V. Their lower MAE can reflect cancellation between systematic contributions and should not be read as superior second-reduction accuracy. Squared bias accounts for approximately 46% of the overall mean-squared error. A molecule-held-out constant offset reduces MAE/RMSE to 0.158/0.211 V; separate first- and second-step offsets give 0.106/0.166 V.

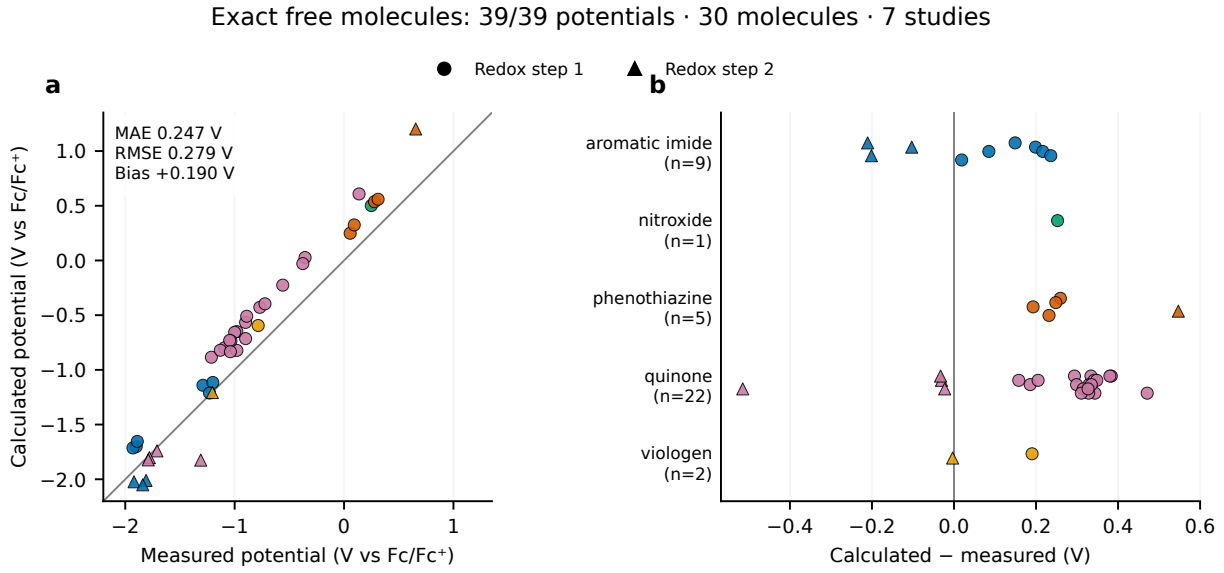


Figure 2: Unfitted retrospective benchmark for 39 exact free-parent potentials in MeCN. Family and redox-step labels expose unequal coverage. All sourced endpoints and outliers are retained; no fitted offset is applied.

Study-held-out values are 0.182/0.232 and 0.124/0.177 V, respectively (SI). These internal analyses diagnose structure in the residuals while leaving the primary predictions unchanged.

The computed Fc absolute potential is 4.379 V. Stored comparison values, 4.618–4.662 V, use other functionals and cannot be adopted as level-matched replacements. Their difference from the present reference is similar to the positive first-step bias, making a reference contribution plausible. Multiple organic-reference differences that cancel Fc are examined in the SI. Nevertheless, all nine paired step spacings are too large in magnitude: a common reference shift cancels from a spacing and cannot explain this result. Reference, charge-state solvation and ion-pairing errors can coexist. A large quinone study found distinct first/second-reduction intercepts under other methods in DMF versus SCE; it provides relevant precedent, not a numerical calibration for MeCN/Fc.[8] In five neutral-SVPD controls, changing both geometry and Hessian basis lowers the first potential by 11.7–35.1 mV when the original anion and Fc are held fixed. Nuclear contributions are 7.8–21.3 meV; geometry-induced active-energy contributions are 0.4–13.8 meV (SI). The imbalance is measurable but does not explain the full observed bias. A spin-only partition also omits ferrocenium orbital/vibronic contributions; their sign and limiting size are discussed in the SI.[28] Changing the nuclear rotor cutoff over 25, 50 and 100 cm^{-1} gives a maximum benchmark endpoint span of 25.4 mV and aggregate MAE values of 0.245–0.247 V (SI). The primary 50 cm^{-1} convention is retained.

3.3 A fixed base shifts grafting thermodynamics without changing ranking

Of 75 local reference reactions, 67 have complete free energies and eight remain held. The base pilot combines five complete candidate reactions with six checked DBU/DIPEA components to give twenty base-assisted values. For a fixed base and ion representation,

$$\Delta G_{\mathrm{graft,B}} = \Delta G_{\mathrm{graft,HCl}} + \Delta G(\mathrm{B} + \mathrm{HCl} \rightarrow \mathrm{BH^+} + \mathrm{Cl^-}), \tag{8}$$

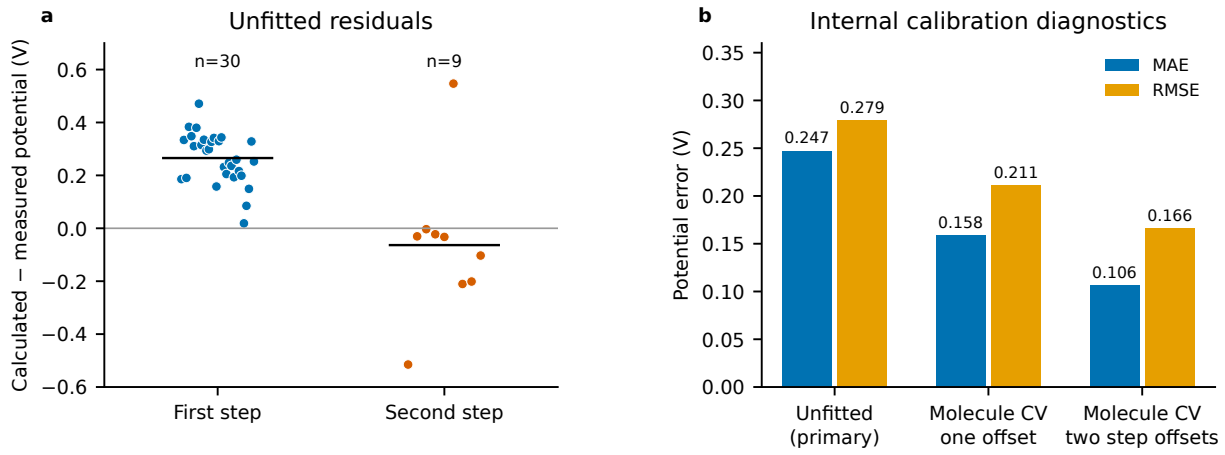


Figure 3: Unfitted step-dependent residuals and internal molecule-held-out offset diagnostics. Training offsets exclude the entire target molecule. Diagnostic calibration is shown separately from the primary benchmark and is not an external test on new compounds.

with the paired analogue and its standard-state correction defined separately. The separated-ion constants are -102.78 kJ mol^{-1} for DBU and -70.98 kJ mol^{-1} for DIPEA. Thus the base choice can change signs but cannot reorder candidates within a given representation.

The two benzoquinone HCl-reference values, $+57.63$ and $+56.03$ kJ mol^{-1}, become $-45.14/-46.75$ kJ mol^{-1} with DBU and $-13.34/-14.94$ kJ mol^{-1} with DIPEA. AQ-1-oxy and NDI also change sign in this panel (Figure 4); this is not a benzoquinone-specific rescue. These values motivate testing a defined base-assisted route, while selectivity, precursor access, salt speciation and support accessibility remain separate route gates.

The revised MeCN collection gives pKaH(DBU)=24.31 and the recommended acid table gives pKa(HCl)=10.30.[23–25] At 298.15 K these imply -79.97 kJ mol^{-1} for capture, versus -102.78 kJ mol^{-1} computed, a discrepancy of -22.81 kJ mol^{-1}. Replacing only this term leaves the two DBU/BQ sums negative ($-22.33/-23.94$ kJ mol^{-1}), but changes shared-error cancellation because HCl cancels from the full DFT cycle. These hybrid sums are sensitivities, not calibrated grafting predictions. Two additional HCl/Cl-free relative-basicity reactions, for pyridine and triethylamine against DBU, differ from the tabulated scale by $+4.58$ and $+1.40$ kJ mol^{-1} (SI). This narrows the diagnostic without uniquely assigning the capture discrepancy to chloride or HCl. Exact source temperature/activity conventions remain under review. Independent experimental DIPEA validation remains unresolved. The reported MeCN value 18.6 carries a specific PCM-estimate note in the author-posted source; it is not admitted as an independent measured anchor.[29] DIPEA remains conditional, and the DBU discrepancy is not transferred as a correction.

All 87 completed reference and base-assisted reaction values retain their nominal sign over those cutoff scenarios; the largest span is 4.46 kJ mol^{-1}(SI). This settings test leaves the independent route-validation requirements unchanged.

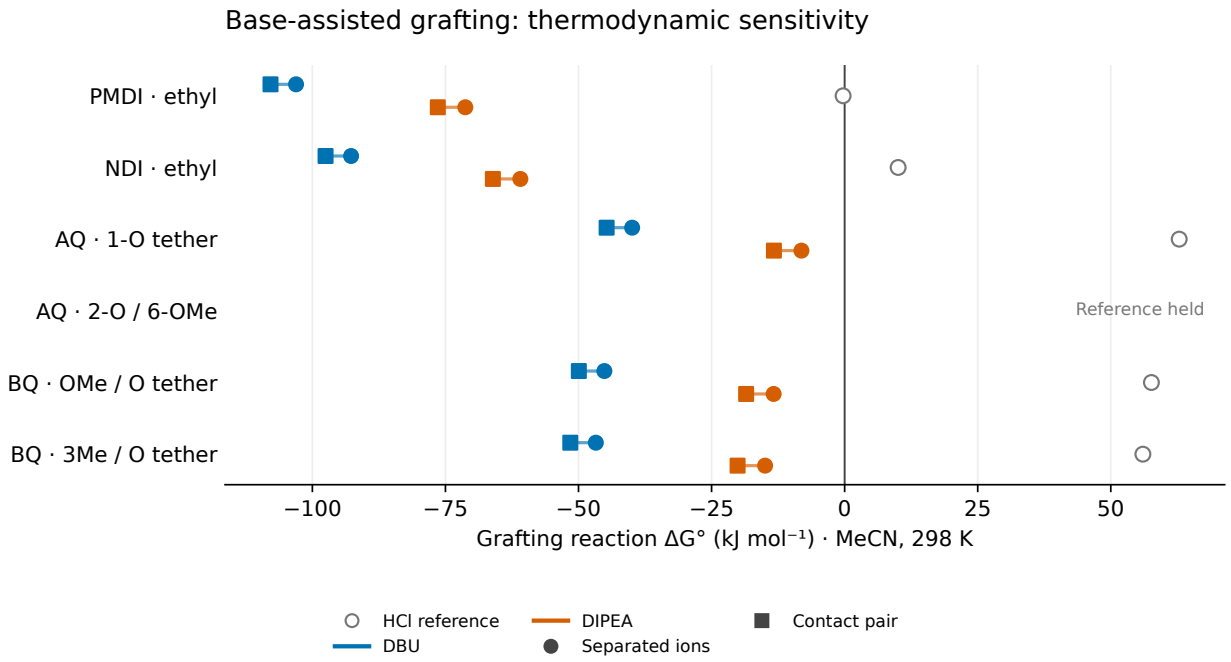


Figure 4: Candidate HCl-reference reactions plus a constant per-base proton-capture term. Separated ions and contact pairs are alternative thermodynamic representations; their differences do not establish measured ion-pair populations.

3.4 An exact PMDI design remains pending after molecular screening

The enumerated PMDI-ethyl design maps exactly to the mono-N-ethyl/NH precursor and its N-(4-methylbenzyl) graft-site model. Its checked molecular potentials are −1.076 and −1.977 V versus the computed Fc reference. The HCl-reference grafting value is −0.29 kJ mol^{-1}; including the defined DBU/separated-ion term gives −103.07 kJ mol^{-1}. These values provide molecular and reaction-model evidence, with the benchmark and proton-transfer limitations described above.

Example 19 of EP0010440B1 reports isolation of the exact mono-N-ethyl/NH PMDI precursor with a stated NMR assignment.[30] Molecular N-alkylation of PMDI in the presence of CsF has also been reported, with crystallographic characterization of an N,N′-diethyl derivative.[31] These reports support precursor access and the attachment reaction class, respectively. They do not establish DBU-assisted benzylation of the exact mono-N-ethyl precursor, selective attachment to a Merrifield-type bead or retention of redox performance after grafting. The exact route therefore remains pending for selectivity and core-retention evidence.

The neutral capped molecule has a nominal two-electron capacity of 153.9 mAh g^{-1}. Retaining scaffold and residual chloride mass changes the composition-based ceiling: at three reactive handles per five repeats, one, two or three PMDI grafts give 61.2, 98.9 or 124.5 mAh g^{-1}, respectively (Figure 5). These are conditional bounds for a dry neutral linear fragment with full utilization of an assumed two-electron budget. Crosslinker, electrolyte and charging counterion mass are excluded unless explicitly added. No graft-conversion distribution or accessible polymer capacity is inferred. An application target exceeding a scenario's ceiling would rule out that scenario; lying below the ceiling would still require independent evidence for utilization, stability and transport.

The exact-candidate occupancy graphs and assessment protocol are specified in the SI; complete matched occupancy thermodynamics and intersite coupling remain unavailable. Thus enumeration,

redox and route evidence are linked to a defensible pending decision, while a complete polymer-level selection remains open.

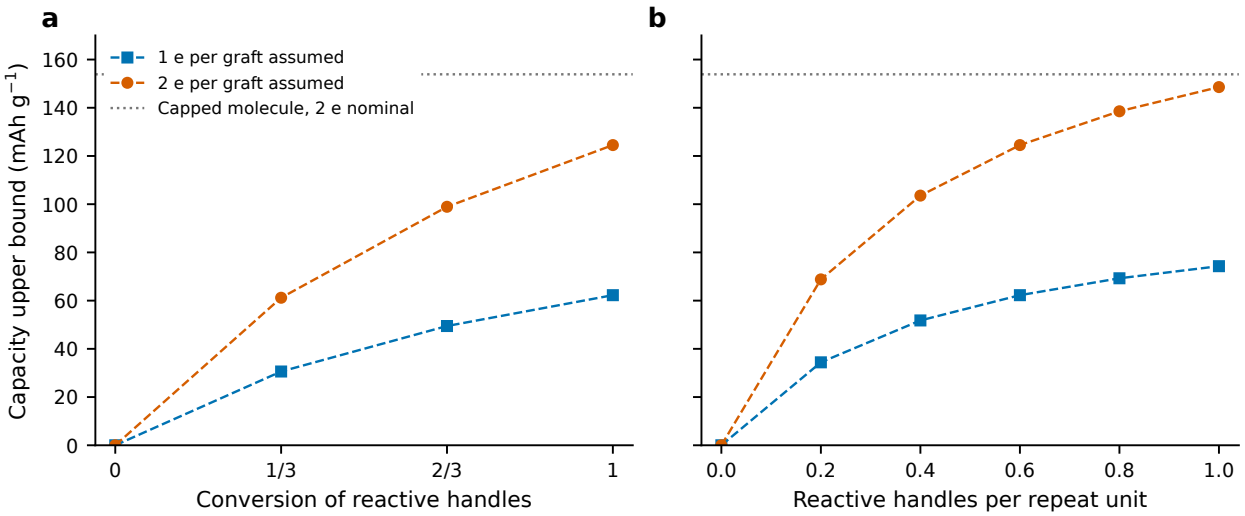


Figure 5: Composition-dependent PMDI capacity ceilings. The neutral H-capped fragment has five repeats; panel (a) fixes three reactive handles and varies conversion, while panel (b) varies handle density at full conversion. One- and two-electron budgets are assumptions. Residual chlorides and scaffold mass are retained. The dotted capped-molecule descriptor excludes the polymer scaffold; none of these curves establishes accessible material capacity.

3.5 Conditional grafting is demonstrated in a finite control

A five-repeat phthalimide model distinguishes reactive sites, plain-styrene end padding and occupied sites. All twelve electronic poses and their twelve native free energies pass their specified checks and independent reconstruction. A Boltzmann partition over the two known poses of each graph gives adjacent and one-spacer nonadditivities of -5.17 and -11.39 kJ mol^{-1}. These values include backbone, conformational and solvent response. They quantify conditional attachment within the declared set, rather than a pure motif-pair interaction.

The sequence interval does not fix spatial separation in these controls (Figure 6). Each occupied graph has both a close-core and a separated-core local minimum. Adjacent-site core centroids are 4.24 and 16.66 Å apart; the corresponding one-spacer distances are 4.77 and 20.20 Å. Within each sampled pair, the close-contact conformer has lower G by 11.9 and 7.1 kJ mol^{-1}, respectively. These same-graph differences include changes throughout the chain and solvent response, distinct from the four-state grafting nonadditivity. The one-spacer model therefore permits motif contact through backbone folding.

The maximum configurational lowering among these six two-pose sets is 0.14 kJ mol^{-1}. This small change describes only the two known conformers; it does not bound the error from undiscovered conformers. Changing the rotor cutoff gives cycle ranges of -5.17 to -4.64 and -12.34 to -11.39 kJ mol^{-1}. Broader sampling, multiple backbone configurations and DFT re-ranking remain necessary before interpreting a spacing design rule. Construction, conditional reactions and the finite-partition protocol are reported in the SI.

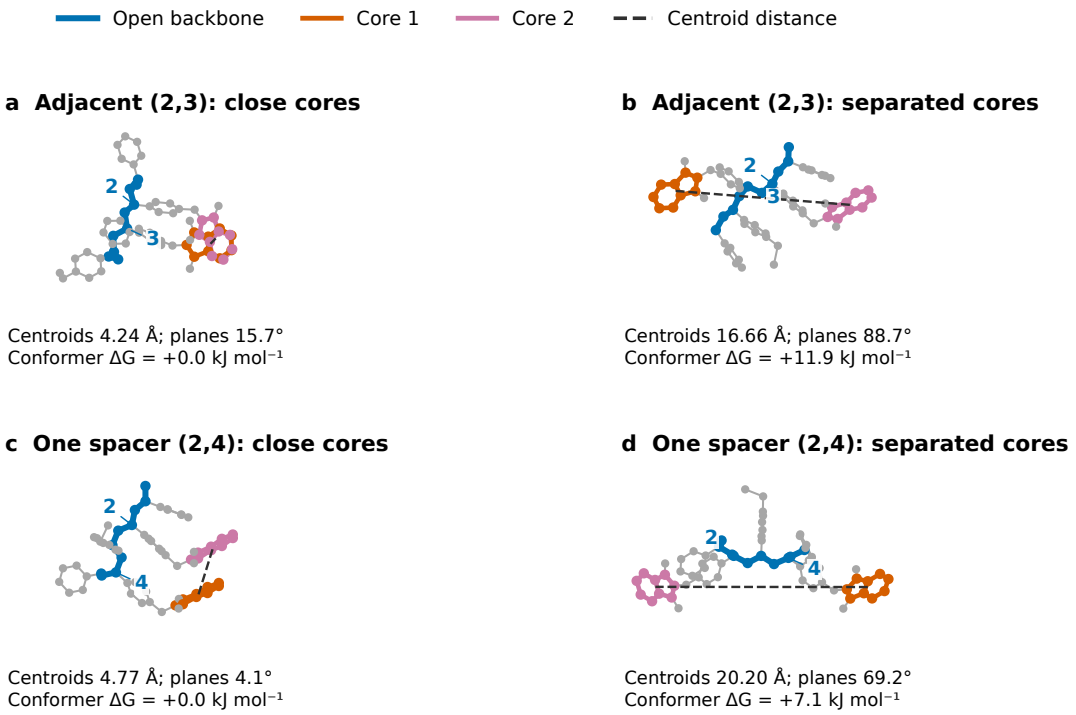


Figure 6: Accepted phthalimide control geometries at adjacent (a,b) and one-spacer (c,d) occupancy. Blue marks the open backbone; numbers identify occupied repeats. Orange and magenta mark conjugated core-ring atoms. Ungrafted site 3 in the one-spacer graph retains benzyl chloride; the end-padding repeats are plain styrene. Orthographic views use the saved XYZ coordinates, proper rotations and a common spatial scale; hydrogens are hidden and bond lines indicate checked connectivity. Dashed lines join core centroids, and angles refer to their fitted planes. Conformer ΔG is relative to the lower G in the same two-pose graph at 298.15 K and a 50 cm^{-1} rotor cutoff. These finite sets do not establish converged populations or isolated core-binding free energies.

4 Discussion

The immediate chemical lesson is that a candidate must satisfy each required dimension within a defined context. The PMDI example shows how precursor access and favorable model thermodynamics can coexist with an unresolved material decision. Base capture contributes a common term rather than a candidate-specific ranking advantage; its external check limits the interpretation of a favorable grafting sum. Likewise, conditional occupancy cycles state what can be computed when graft conversion and spacing are unknown, without inventing a loading distribution. The finite controls show why sequence spacing and spatial contact require separate assessment: inserting an ungrafted repeat still permits a close-contact local minimum. Whole-chain conformer free energies and conditional grafting cycles describe complementary responses. Electronic coupling and full-bead transport require additional models.

The benchmark identifies a sizeable common bias and a separate step-dependent component. Reference cancellation, grouped cross-validation and paired spacings offer complementary diagnostics; none alone attributes error uniquely to Fc. Changing the fitted offset or charge-state protocol now would require a declared calibration set and a separate external test before claiming improved predictive accuracy.

The scope of the evidence determines how far these assessments transfer. The free-parent benchmark is concentrated in quinones and provides limited support for diimides and viologens. The basis controls quantify a specific protocol sensitivity rather than establish basis convergence. Base-assisted sums remain sensitive to continuum solvation and ion speciation, while the finite occupancy partition does not bound contributions from undiscovered conformers or an atactic polymer ensemble. For an exact candidate, these uncertainties accompany the chemical requirements for selective attachment, redox-core retention and accessible utilization. An all-criterion decision therefore requires candidate-specific evidence under a consistent set of operating and reaction conditions.

5 Conclusions

Molecular electrochemistry, attachment thermodynamics and neighboring-site behavior provide complementary evidence for selecting redox motifs on preformed polymers. The retrospective benchmark identifies both common and redox-step-dependent errors. Balanced grafting cycles show that a fixed base shifts reaction free energies without changing within-base order, and matched occupancy cycles quantify how a neighboring graft changes attachment thermodynamics within a specified conformer set. Spatial analysis of the same controls shows that sequence-separated grafts can still form close contacts. The exact PMDI example further shows that including scaffold mass changes the interpretation of a molecular capacity descriptor. These findings support a route-aware assessment in which all required criteria must pass independently; favorable molecular or reaction values alone leave a polymer candidate unresolved when its chemical and material requirements remain untested.

Supporting Information

Benchmark intervals and grouped offset diagnostics; relative-basicity controls; finite-polymer construction and conditional cycles; electronic fragment decomposition; numerical implementation, concurrency and quantum restart controls; evidence terminology.

Data and code availability

The immutable evidence epoch and source hashes are recorded under `manuscript/evidence/2026-10-24_spatial_revision/`. Versioned code and curated metadata are maintained in the public project repository. A locally prepared reviewer package contains the raw geometries, energies, frequencies and state metadata used for the primary calculations and auxiliary sensitivity controls. Its offline replay reconstructs the primary benchmark, grafting and finite-ensemble values after relocation without opening original cluster paths; package scope, environment records and integrity checks are described in the Supporting Information. A separately pinned CUDA 12.8 compatibility profile has been installed afresh and reproduces complete singlet and doublet quantum controls within predeclared tolerances (Supporting Information). This verifies a tested Linux/CUDA deployment rather than unrestricted platform portability. The raw-state package does not yet have a public archival DOI; final deposition metadata remain outstanding.

Author contributions, funding and competing interests

These statements require confirmation by the authors and are intentionally not inferred in this working draft.

References

[1] Tobias Janoschka, Norbert Martin, Udo Martin, Christian Friebe, Sabine Morgenstern, Hannes Hiller, Martin D. Hager, and Ulrich S. Schubert. An aqueous, polymer-based redox-flow battery using non-corrosive, safe, and low-cost materials. *Nature*, 527:78–81, 2015. doi: 10.1038/nature15746.

[2] Elena C. Montoto, Gavvalapalli Nagarjuna, Jingshu Hui, Mark Burgess, Nina M. Sekerak, Kenneth Hernández-Burgos, Teng-Sing Wei, Marissa Kneer, Joshua Grolman, Kevin J. Cheng, Jennifer A. Lewis, Jeffrey S. Moore, and Joaquín Rodríguez-López. Redox Active Colloids as Discrete Energy Storage Carriers. *Journal of the American Chemical Society*, 138:13230–13237, 2016. doi: 10.1021/jacs.6b06365.

[3] Dukhan Kim, Melanie S. Sanford, Thomas P. Vaid, and Anne J. McNeil. A nonaqueous redox-matched flow battery with charge storage in insoluble polymer beads. *Chemistry—A European Journal*, 28:e202200149, 2022. doi: 10.1002/chem.202200149.

[4] Süleyman Er, Changwon Suh, Michael P. Marshak, and Alán Aspuru-Guzik. Computational design of molecules for an all-quinone redox flow battery. *Chemical Science*, 6:885–893, 2015. doi: 10.1039/c4sc03030c.

[5] Rebekah Duke, Vinayak Bhat, Parker Sornberger, Susan A. Odom, and Chad Risko. Towards a comprehensive data infrastructure for redox-active organic molecules targeting non-aqueous redox flow batteries. *Digital Discovery*, 2:1152–1162, 2023. doi: 10.1039/d3dd00081h.

[6] Aleksandr V. Marenich, Junming Ho, Michelle L. Coote, Christopher J. Cramer, and Donald G. Truhlar. Computational electrochemistry: prediction of liquid-phase reduction potentials. *Phys. Chem. Chem. Phys.*, 16:15068–15106, 2014. doi: 10.1039/c4cp01572j.

[7] Mansoor Namazian, Ching Yeh Lin, and Michelle L. Coote. Benchmark Calculations of Absolute Reduction Potential of Ferricinium/Ferrocene Couple in Nonaqueous Solutions. *Journal of Chemical Theory and Computation*, 6:2721–2725, 2010. doi: 10.1021/ct1003252.

[8] Sarah Elhajj and Samer Gozem. First and Second Reductions in an Aprotic Solvent: Comparing Computational and Experimental One-Electron Reduction Potentials for 345 Quinones. *Journal of Chemical Theory and Computation*, 20:6227–6240, 2024. doi: 10.1021/acs.jctc.4c00602.

[9] Cedrik Wiberg, Lars Evenäs, Michael Busch, and Elisabet Ahlberg. Naphthalene diimides (NDI) in highly stable ph-neutral aqueous organic redox flow batteries. *Journal of Electroanalytical Chemistry*, 896:115224, 2021. doi: 10.1016/j.jelechem.2021.115224.

[10] Sebastiaan P. Huber, Spyros Zoupanos, Martin Uhrin, Leopold Talirz, Leonid Kahle, Rico Häuselmann, Dominik Gresch, Tiziano Müller, Aliaksandr V. Yakutovich, Casper W. Andersen, Francisco F. Ramirez, Carl S. Adorf, Fernando Gargiulo, Snehal Kumbhar, Elsa Passaro, Conrad Johnston, Andrius Merkys, Andrea Cepellotti, Nicolas Mounet, Nicola Marzari, Boris Kozinsky, and Giovanni Pizzi. AiiDA 1.0, a scalable computational infrastructure for automated reproducible workflows and data provenance. *Scientific Data*, 7:300, 2020. doi: 10.1038/s41597-020-00638-4.

[11] Anubhav Jain, Shyue Ping Ong, Wei Chen, Bharat Medasani, Xiaohui Qu, Michael Kocher, Miriam Brafman, Guido Petretto, Gian-Marco Rignanese, Geoffroy Hautier, Daniel Gunter, and Kristin A. Persson. FireWorks: a dynamic workflow system designed for high-throughput applications. *Concurrency and Computation: Practice and Experience*, 27:5037–5059, 2015. doi: 10.1002/cpe.3505.

[12] James W. Furness, Aaron D. Kaplan, Jinliang Ning, John P. Perdew, and Jianwei Sun. Accurate and numerically efficient r2SCAN meta-generalized gradient approximation. *The Journal of Physical Chemistry Letters*, 11:8208–8215, 2020. doi: 10.1021/acs.jpclett.0c02405.

[13] Eike Caldeweyher, Sebastian Ehlert, Andreas Hansen, Hagen Neugebauer, Sebastian Spicher, Christoph Bannwarth, and Stefan Grimme. A generally applicable atomic-charge dependent London dispersion correction. *The Journal of Chemical Physics*, 150:154122, 2019. doi: 10.1063/1.5090222.

[14] Aleksandr V. Marenich, Christopher J. Cramer, and Donald G. Truhlar. Universal solvation model based on solute electron density and on a continuum model of the solvent defined by the bulk dielectric constant and atomic surface tensions. *The Journal of Physical Chemistry B*, 113:6378–6396, 2009. doi: 10.1021/jp810292n.

[15] Narbe Mardirossian and Martin Head-Gordon. ωB97M-V: A combinatorially optimized, range-separated hybrid, meta-GGA density functional with VV10 nonlocal correlation. *The Journal of Chemical Physics*, 144:214110, 2016. doi: 10.1063/1.4952647.

[16] Florian Weigend and Reinhart Ahlrichs. Balanced basis sets of split valence, triple zeta valence and quadruple zeta valence quality for H to Rn: Design and assessment of accuracy. *Physical Chemistry Chemical Physics*, 7:3297–3305, 2005. doi: 10.1039/B508541A.

[17] Dmitrij Rappoport and Filipp Furche. Property-optimized Gaussian basis sets for molecular response calculations. *The Journal of Chemical Physics*, 133:134105, 2010. doi: 10.1063/1.3484283.

[18] Qiming Sun et al. Recent developments in the PySCF program package. *The Journal of Chemical Physics*, 153:024109, 2020. doi: 10.1063/5.0006074.

[19] Rui Li, Qiming Sun, Xing Zhang, and Garnet Kin-Lic Chan. Introducing GPU-acceleration into the Python-based simulations of chemistry framework, 2024. Preprint.

[20] Stefan Grimme. Supramolecular Binding Thermodynamics by Dispersion-Corrected Density Functional Theory. *Chemistry – A European Journal*, 18:9955–9964, 2012. doi: 10.1002/chem.201200497.

[21] Christoph Bannwarth, Eike Caldeweyher, Sebastian Ehlert, Andreas Hansen, Philipp Pracht, Jakob Seibert, Sebastian Spicher, and Stefan Grimme. Extended tight-binding quantum chemistry methods. *WIREs Computational Molecular Science*, 11:e1493, 2021. doi: 10.1002/wcms.1493.

[22] Lee-Ping Wang and Chenchen Song. Geometry optimization made simple with translation and rotation coordinates. *The Journal of Chemical Physics*, 144:214108, 2016. doi: 10.1063/1.4952956.

[23] Sofja Tshepelevitsh, Agnes Kütt, Märt Lõkov, Ivari Kaljurand, Jaan Saame, Agnes Heering, Paul G. Plieger, Robert Vianello, and Ivo Leito. On the basicity of organic bases in different media. *European Journal of Organic Chemistry*, 2019:6735–6748, 2019. doi: 10.1002/ejoc.201900956.

[24] Elin Raamat, Karl Kaupmees, Gea Ovsjannikov, Aleksander Trummal, Agnes Kütt, Jaan Saame, Ivar Koppel, Ivari Kaljurand, Lauri Lipping, Toomas Rodima, Viljar Pihl, Ilmar A. Koppel, and Ivo Leito. Acidities of strong neutral Brønsted acids in different media. *Journal of Physical Organic Chemistry*, 26:162–170, 2013. doi: 10.1002/poc.2946.

[25] University of Tartu, Chair of Analytical Chemistry. Author-maintained basicity collection and recommended acid table. https://analytical.chem.ut.ee/databases/pka-values/. Accessed 22 October 2026. Acid table: https://analytical.chem.ut.ee/HA_UT/pKa_Table_Water_Acetonitrile_DCE_Gas_Phase.pdf.

[26] Christoph Bannwarth, Sebastian Ehlert, and Stefan Grimme. GFN2-xTB—an accurate and broadly parametrized self-consistent tight-binding quantum chemical method with multipole electrostatics and density-dependent dispersion contributions. *Journal of Chemical Theory and Computation*, 15:1652–1671, 2019. doi: 10.1021/acs.jctc.8b01176.

[27] Sebastian Ehlert, Marcel Stahn, Sebastian Spicher, and Stefan Grimme. Robust and efficient implicit solvation model for fast semiempirical methods. *Journal of Chemical Theory and Computation*, 17:4250–4261, 2021. doi: 10.1021/acs.jctc.1c00471.

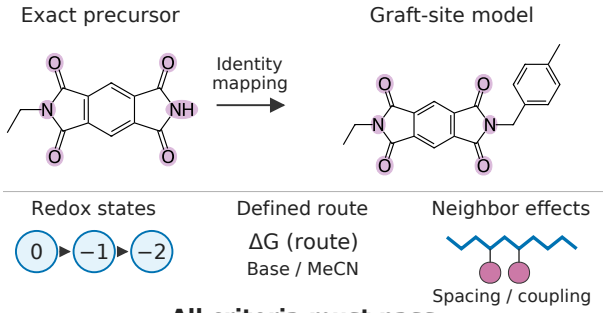
[28] Jan Fulara, Karol Filipkowski, and John P. Maier. Electronic Transition of Ferrocenium: Neon Matrix and CASPT2 Studies. *The Journal of Physical Chemistry C*, 121:10694–10697, 2017. doi: 10.1021/acs.jpcc.6b10391.

[29] David J. Heldebrant, Phillip K. Koech, James E. Rainbolt, Feng Zheng, Tricia Smurthwaite, Charles J. Freeman, Merit Oss, and Ivo Leito. Performance of single-component CO2-binding organic liquids (CO2BOLs) for post combustion CO2 capture. *Chemical Engineering Journal*, 171(3):794–800, 2011. doi: 10.1016/j.cej.2011.02.012.

[30] Richard J. Bochis, Michael H. Fisher, and Bruce O. Linn. Pyromellitic diimides, processes for their preparation, these compounds in orally administrable form for use in increasing the feed efficiency of a ruminant animal and compositions for this purpose containing them. European patent EP0010440B1, published January 26, 1983, 1983. URL https://patents.google.com/patent/EP0010440B1/en. Example 19, printed p. 10; precursor identity only.

[31] Darren G. Hamilton, Daniel E. Lynch, Karl A. Byriel, Colin H. L. Kennard, and Jeremy K. M. Sanders. N-alkylation of pyromellitic diimide: Solid-state structure of the 1 : 1 N,N′-diethylpyromellitic diimide/2,6-dimethoxynaphthalene cocrystal. *Australian Journal of Chemistry*, 51(5):441–444, 1998. doi: 10.1071/C97214.

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